

# Lukas Gosch

18/5/1993 | Citizenship: Austrian

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My research focuses on the intersection of Machine Learning (ML) with optimization. I have done extensive work on graph neural networks and the use of concepts from mathematical programming to prove properties such as robustness. Currently, I'm interested in ML for optimization, e.g., how to speed up or learn solving optimization problems, and how predictions and optimization can be effectively combined.

## EDUCATION

### PHD MACHINE LEARNING

TECHNICAL UNIVERSITY OF MUNICH

January 2022 - July 2026 (expected graduation)

- Supervised by Prof. Stephan Günnemann
- Part of the Konrad Zuse School of Excellence in Reliable AI

### MACHINE LEARNING SUMMER SCHOOL (MLSS)

March 2024 - Okinawa Institute of Science and Technology (OIST) & RIKEN AIP, Japan

### MSC COMPUTATIONAL SCIENCE

UNIVERSITY OF VIENNA

March 2018 - July 2021

- Focus: Machine Learning and Optimization
- Master's thesis award in Operations Research
- GPA: 1.0 (1 best, 5 worst)

### BSC TECHNICAL PHYSICS

TECHNICAL UNIVERSITY OF VIENNA

September 2013 - September 2017

- Thesis: C++ data analysis in high energy physics
- GPA: 1.9 (1 best, 5 worst)

## ACADEMIC HONORS & AWARDS

### BEST PAPER AWARD

AdvML-Frontiers @ NeurIPS 2024

### BEST MASTER'S THESIS PRICE

Awarded by the Austrian Society of Operations Research (ÖGOR), 2021

### DEAN'S LIST AWARD

University of Vienna - Faculty of Physics, 2021

### PERFORMANCE SCHOLARSHIP

University of Vienna, 2020

## TEACHING

### TECHNICAL UNIVERSITY OF MUNICH

MASTER THESES SUPERVISED

- Morgane Ayle, Francesco Campi, Daniel Sturm, Philipp Foth, Ajinkya Mohgaonkar (ongoing)

GUIDED RESEARCH SUPERVISED

- Xichuan Chen (ongoing)

COURSE INSTRUCTOR

- Machine Learning
- ML for Graphs & Sequential Data
- Seminar: Selected Topics in ML Research

## WORK EXPERIENCES

### TECHNICAL UNIVERSITY OF MUNICH

PHD STUDENT, DATA ANALYTICS AND MACHINE LEARNING (DAML) GROUP

January 2022 - now | Munich, Germany

Research Topics:

- Graph Neural Networks
- Adversarial Robustness and Certification
- Machine Learning for Optimization

Published at main ML conferences such as NeurIPS and ICLR. My works got featured for spotlight presentation and won a best paper award.

### AIT AUSTRIAN INSTITUTE OF TECHNOLOGY

MASTER THESIS, ENERGY LOGISTICS & TRANSPORT OPTIMIZATION GROUP

July 2020 - June 2021 | Vienna, Austria

Combinatorial optimization applied in green logistics

- Mathematical modelling introducing a novel network design problem
- Exact and heuristic solving using integer linear programming and hybrid heuristics (C++)
- Published at the IPIC2021 conference

### INSTITUTE OF SCIENCE & TECHNOLOGY AUSTRIA (ISTA)

RESEARCH INTERNSHIP, MACHINE LEARNING AND COMPUTER VISION GROUP

October 2019 - April 2020 | Vienna, Austria

Supervised by Prof. Christoph Lampert | Technical Report

- Information-theoretic approaches for deep self-supervised representation learning

### FRAUNHOFER INSTITUTE FOR ALGORITHMS & SCIENTIFIC COMPUTING

RESEARCH INTERNSHIP, NUMERICAL DATA DRIVEN PREDICTION RESEARCH GROUP

July 2018 - September 2018 | Bonn, Germany

Python - Topological data analysis applied to time series

### QUEEN'S UNIVERSITY BELFAST

RESEARCH INTERNSHIP, SCHOOL OF PSYCHOLOGY

July 2017 - August 2017 | Northern Ireland, UK

Matlab - Ball trajectory reconstruction from 3D markers

2008 - 2018 | Further employments

## PUBLICATIONS

\* ... Equal Contribution

### PUBLICATION

- M. Sabanayagam\*, **L. Gosch**\*, S. Günnemann, D. Ghoshdastidar "Exact Certification of (Graph) Neural Networks Against Label Poisoning" Spotlight @ ICLR 2025 - top 5.1%
- **L. Gosch**\*, M. Sabanayagam\*, D. Ghoshdastidar, S. Günnemann "Provable Robustness of (Graph) Neural Networks Against Data Poisoning and Backdoor Attacks" Best Paper Award - AdvML @ NeurIPS 2024
- M. Kollovich, **L. Gosch**, Y. Scholten, M. Lienen, S. Günnemann "Assesing Robustness via Score-based Adversarial Image Generation" TMLR 2024
- P. Foth, **L. Gosch**, S. Geisler, L. Schwinn, S. Günnemann "Relaxing Graph Transformers for Adversarial Attacks" Differentiable Almost Everything @ ICML 2024
- **L. Gosch**\*, S. Geisler\*, D. Sturm\*, B. Charpentier, D. Zügner, S. Günnemann "Adversarial Training for Graph Neural Networks: Pitfalls, Solutions, and New Directions" NeurIPS 2023
- F. Campi, **L. Gosch**, T. Wollschläger, Y. Scholten, S. Günnemann "Expressivity of Graph Neural Networks Through the Lens of Adversarial Robustness" AdvML @ ICML 2023
- **L. Gosch**, D. Sturm, S. Geisler, S. Günnemann "Revisiting Robustness in Graph Machine Learning" ICLR 2023, also TSRML & ML Safety @ NeurIPS 2022
- M. Ayle, J. Schuchardt, **L. Gosch**, D. Zügner, S. Günnemann "Training Differentially Private Graph Neural Networks with Random Walk Sampling", TSRML@NeurIPS 2022
- **L. Gosch**, M. Prandtstetter, K. F. Doerner "On Modelling and Solving Green Collaborative Transportation Planning" IPIC2021
- R. Feldbauer, **L. Gosch**, L. Lüftinger, P. Hyden, A. Flexer, T. Rattei "DeepNOG: fast and accurate protein orthologous group assignment" Bioinformatics 2020

## SOFTWARE

### DEEPNOG

Tool for protein similarity search using deep networks. Published in Bioinformatics.

## SKILLS

- Python (PyTorch, PyG, NumPy, scikit-learn)
- Gurobi, CPLEX, C++,  $\LaTeX$  GIT, Linux
- Some Experiences in C, Java, R, Matlab, SQL
- German (Native), English (C1, IELTS 8.5)

## FURTHER ACTIVITIES

### INVITED TALKS

- INSAIT, Sofia, 2024
- Adv-ML @ NeurIPS 2024 (Oral Presentation)
- MDSI InterConnect, TUM, 2024
- Theoretical Foundations of AI Group, TUM, 2023
- TSRML @ NeurIPS 2022 (Contributed Talk)

### REVIEWER

- ICML 2025
- ICLR 2024
- Differentiable Almost Everything@ICML 2024
- SynS & ML @ ICML 2023

### STUDENT REPRESENTATIVE

KONRAD ZUSE SCHOOL OF EXCELLENCE IN RELIABLE AI (GRADUATE SCHOOL)  
2022 - 2023

- Organization of Social Events, Journal Club

### BOOK CLUBS

DAML READING GROUP | 2022 - 2024

- Read "High-Dimensional Statistics: A Non-Asymptotic Viewpoint" by M. Wainwright
- Read "The Theory of Statistics and Its Applications" by D. Cox

SCIENTIFIC WRITING | 2019 - 2020

- With colleagues from IST Austria

CLASSIC LITERATURE & PHILOSOPHY | 2017 - 2021

- Founder
- Book list: <https://saper0.github.io/lesekreis/>

### ATTENDING OF LECTURES AND EVENTS

ON PHILOSOPHY, ETHICS, AND MACHINE LEARNING  
Examples

- Inquire - Ethic conversations at the LMU
- Philosophy courses - University of Vienna

## REFERENCES

References Upon Request.